Accessing the symmetric component of quadrupole misalignment through counter-rotating closed orbits in EDM storage rings

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In the proposed proton electric dipole moment (EDM) storage ring, micron-level quadrupole misalignments mimic an EDM through a geometric phase. Splitting the misalignment of each focusing–defocusing pair into a symmetric and an antisymmetric component, we show that the symmetric one distorts the closed orbit four to seven times more weakly at low harmonics, so single-beam orbit correction leaves it largely in place. With symplectic beam and spin tracking at the frozen-spin energy, we show that a joint fit of the two counter-rotating orbits recovers it from their sum. Because static monitor offsets are degenerate with this component, the scheme monitors drifts relative to a beam-based-alignment reference. We determine its domain of validity against monitor motion, electrostatic deflector errors, corrector type and correlated misalignment, and follow it through a one-week scenario. Under the conditions studied, two-beam monitoring keeps the false EDM well below target between alignments without quadrupole polarity reversal.

I. INTRODUCTION

A. Frozen spin method and the false EDM

A permanent electric dipole moment of an elementary particle violates parity and time reversal and, assuming CPT invariance, CP. The Standard Model predicts about 10^{-31} e·cm for the proton EDM. The proposed storage-ring experiment [1, 2] stores protons at the magic momentum ($p \approx 0.7$ GeV/c), where the spin follows the momentum in the horizontal plane. An EDM couples to the radial electric field of the deflectors and tilts the spin out of the plane at the rate

$$f \equiv dS_y/dt, \qquad d = 10^{-29}\, e\cdot\text{cm} \longleftrightarrow f \approx 1\ \text{nrad/s}, \quad (1)$$

measured by polarimetry [3]. Couplings of the magnetic moment that mimic this rate are false EDM signals; here we study those of misaligned quadrupoles.

A misaligned quadrupole exposes the beam to a dipole field proportional to the misalignment. A vertical misalignment rotates the spin in the vertical plane, a horizontal one in the horizontal plane. Each rotation alone averages out along the closed orbit, but rotations about orthogonal axes do not commute, and a net out-of-plane rotation per turn remains. This geometric (Berry) phase [4–7] makes the false EDM bilinear in the horizontal and vertical misalignments,

$$f_{\text{false}} \propto dx \cdot dy \quad \rightarrow \quad |f_{\text{false}}| \propto \sigma^2, \qquad (2)$$

with σ the rms misalignment. At $\sigma = 10$ μm it is 10–10^3 times the target, depending on the pattern.

B. Ring design

We use the symmetric-hybrid ring of Ref. [2]: 24 FODO cells of electrostatic deflectors (radius 95.5 m, $E_0 = 4.4$ MV/m) and magnetic quadrupoles of equal gradient magnitude ($g = 0.2$ T/m) (Fig. 1, Table I).

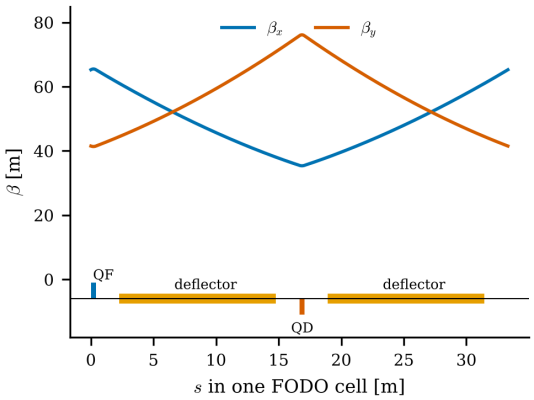


FIG. 1. One of the 24 FODO cells: layout and beta functions. The vertical focusing of the deflectors is zero ($n = 1$); their horizontal focusing is included via the measured tune.

TABLE I. Ring and beam parameters, consistent with Ref. [2].

Circumference / bending radius	800 m / 95.5 m
Momentum; β, γ	0.7007 GeV/c; 0.598, 1.248
Revolution time	4.46 μs
Tunes (Q_x, Q_y)	(2.64, 2.3)
β_x at QF / QD	65 m / 35 m
β_y at QF / QD	42 m / 76 m
$D^{\max}$; slip factor	34 m; −0.25

C. Symmetric and antisymmetric components

The quadrupoles are indexed $n = 0, \ldots, 47$; cell $k = 0, \ldots, 23$ contains QF $n = 2k$ and QD $n = 2k + 1$. For

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the misalignment $v[n]$ in one plane,

$$v^s[2k] = v^s[2k+1] = \tfrac{1}{2}\left(v[2k] + v[2k+1]\right),$$
$$v^a[2k] = -v^a[2k+1] = \tfrac{1}{2}\left(v[2k] - v[2k+1]\right), \quad (3)$$

so that $v[n] = v^s[n] + v^a[n]$. The symmetric component displaces QF and QD of a cell in the same direction, the antisymmetric one in opposite directions. The projections P_s and $P_a = I - P_s$ are orthogonal, each spanning 24 dimensions; they are applied to each plane.

D. Correction schemes and contribution

Reference [2] corrects the misalignment by beam-based alignment (BBA), in which each quadrupole gradient is varied and the beam steered until the orbit no longer responds [8, 9], followed by a cancellation that combines clockwise (CW) and counterclockwise (CCW) beams at both quadrupole polarities. Orbit correction based on the singular value decomposition (SVD) of the response matrix [10–13] is continuous and noninvasive, but we show that with one beam it leaves most of the symmetric component (Sec. III), while a joint fit of the two counter-rotating beams recovers it (Sec. IV B). Section V determines where the scheme works, and Sec. VI follows it through a week of operation.

II. SIMULATION METHOD

A single particle and its spin are tracked with a Gauss–Legendre symplectic integrator [14] and the Thomas–BMT equation [15]. The deflector field is $E_r = E_0 R_0/R$ inside and zero outside (hard edge). With an EDM of 10^{-29} e·cm the code gives $dS_y/dt = 9.8 \times 10^{-10}$ rad/s with opposite signs for the two beam directions; false-EDM results are obtained with the EDM coupling off. The orbit response matrices R (48×48 per plane and beam direction) are obtained by tracking. Because the quadrupole kick reverses with the beam direction, $R_{\mathrm{CW}} \approx -R_{\mathrm{CCW}}$ (column correlation -0.95; the electric bending does not reverse).

False EDM at the frozen-spin energy. The particle is launched on the four-dimensional closed orbit, found by minimizing the turn-by-turn position variance, with the spin longitudinal; $S_y(t) = a + bt + \sum_k [c_k \cos\omega_k t + d_k \sin\omega_k t]$ is fitted and the secular slope b kept. The closed orbit fixes position and angle but not energy, and f is very sensitive to it: at $\sigma = 10$ μm a relative momentum change $\delta = 10^{-7}$ changes f by $\approx 290\times$ the target, and the horizontal spin precession rate by $d\omega_h/d\delta = -5.24 \times 10^6$ rad/s, independent of the misalignment. In a misaligned lattice the energy where the horizontal spin is frozen ($\omega_h = 0$) shifts from the design value by an amount proportional to the misalignment ($\sim 10^{-7}$ here). The experiment holds $\omega_h = 0$ by feedback on the measured horizontal spin phase [16]; we use the same reference, measuring f and ω_h at two energies and interpolating to $\omega_h = 0$ (both are linear in δ; a symmetric momentum spread gives the same mean). A launch at fixed $|p| = p_{\mathrm{magic}}$ instead depends on the launch point, because with hard-edge deflectors the momentum varies along a distorted orbit by $\sim 10^{-7}$: shifting the launch point of one pattern by 6, 12 and 18 cells changes the fixed-momentum result from 1088 to 2729, -456 and 95 times the target, while the frozen-spin result stays at 81.9–82.0 (Appendix A).

Benchmarks. Aligned quadrupoles give zero within the numerical floor ($\approx 0.008\times$ target with the scan settings). A uniform roll ϕ of all deflectors gives $|f| = G\gamma \cdot 2\pi\phi/T_{\mathrm{rev}}$ (Sec. V B) to 10^{-5}. Random quadrupole misalignments give $|f| \propto \sigma^p$, $p = 2.0002$ over 2.5–10 μm. This test does not validate the absolute value: a wrong-energy evaluation also scales as σ^2, since both the energy offset and $df/d\delta$ are proportional to σ.

III. FALSE EDM MECHANISM

Since QF and QD have opposite gradients, an antisymmetric pattern kicks the beam in the same direction at both quadrupoles of a cell, a symmetric pattern in opposite directions. In the smooth-focusing picture the symmetric pattern is a pure harmonic $k = 24$ with an orbit response $G_k \propto 1/|Q^2 - k^2|$ about 100 times below that of low harmonics. The measured response differs [Fig. 2(a)]: for a pattern of cell harmonic m the symmetric-to-antisymmetric orbit gain is 0.15–0.25 for $m \le 3$, rising to one at $m = 12$. The orbit response to a kick is weighted by $\sqrt{\beta}$, which differs strongly between QF and QD; the net kick of a symmetric cell, relative to an antisymmetric one, is $|\sqrt{\beta_F} - \sqrt{\beta_D}|/(\sqrt{\beta_F} + \sqrt{\beta_D}) \approx 0.15$ in both planes, as measured. In the SVD of R [Fig. 2(b)] the smallest singular values belong to predominantly symmetric directions; a truncation at 10^{-2} removes 17 (horizontal) and 24 (vertical) of 48 single-beam directions.

The false EDM is a bilinear functional, $f = \sum_{ij} W_{ij} v_{x,i} v_{y,j}$, and splits into four channels $f_{ss} + f_{sa} + f_{as} + f_{aa}$. The antisymmetric content dominates the false EDM as it dominates the orbit, but by a smaller factor: relative to its orbit visibility, the symmetric component is the more harmful one.

IV. CORRECTION SCHEMES

A. Beam-based alignment and single-beam correction

BBA aligns quadrupoles individually and treats both components alike. With $\approx 1\%$ gradient errors, varying one gradient changes the optics of the whole ring ("breathing", Appendix B) and biases the null in proportion to the orbit distortion; iterating removes the bias. In

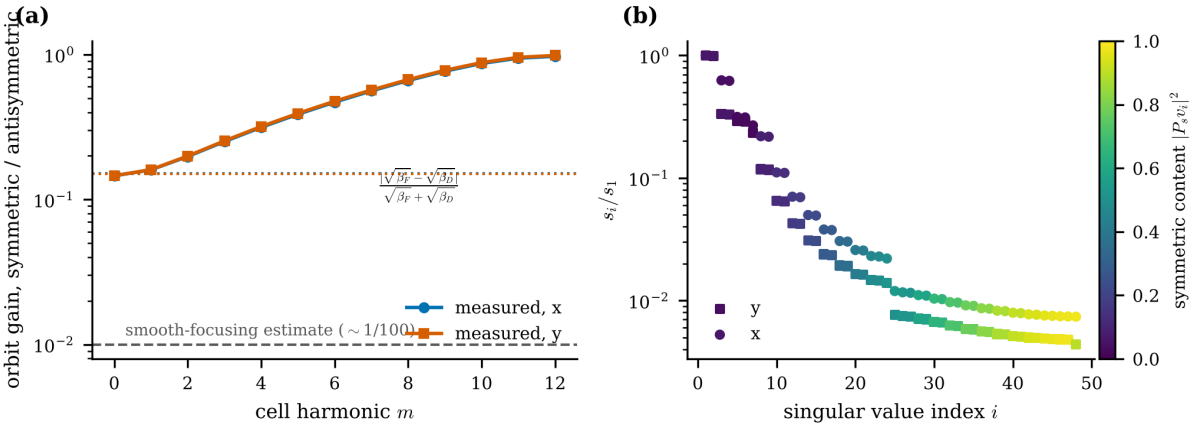


FIG. 2. (a) Symmetric-to-antisymmetric closed-orbit gain for a misalignment pattern of cell harmonic m (tracked matrices). Dotted: $\sqrt{\beta}$ estimate; dashed: smooth focusing. (b) Normalized singular values of the single-beam response matrix, colored by the symmetric content of the right singular vector.

our simulation eight passes reduced the symmetric misalignment from 4.3 (3.3) μm to 0.9 (0.06) μm horizontally (vertically).

Single-beam correction estimates $\hat{v} = R^{+}y$ with a truncated-SVD pseudo-inverse. The symmetric directions have the smallest singular values and are the first truncated, so the result depends on the threshold: for a 5 μm symmetric drift a threshold of 10^{-2} leaves 3.9 μm, 3×10^{-3} leaves 0.05 μm. We choose the threshold by the discrepancy principle—the largest threshold for which the fit residual reaches the BPM noise level—which needs no knowledge of the true misalignment, comes within a factor 1.7 of the best threshold for every estimator used here, and fails to reach the noise level when the measurement model is wrong. Gradient modulation removes BPM offsets but leads to an ill-conditioned difference matrix (Appendix B).

B. Two-beam orbit correction

Both beams are read by the same BPMs, separated by bunch timing. The antisymmetric orbit is nearly opposite for the two beams and the symmetric one nearly equal, up to a phase shift from the reversed order of the two quadrupoles of a cell (Fig. 3). In the vertical plane (Frobenius norms) the antisymmetric component drives the difference channel ($\frac{1}{2}\|(R_{\mathrm{CW}} - R_{\mathrm{CCW}})P_a\| = 45$) and weakly the sum (6.8); the symmetric one appears in both (6.9 and 6.5). In the sum the two components are equally visible, while in one beam the symmetric one is buried under a sevenfold larger antisymmetric response (Appendix C).

Algorithm. (1) After BBA record the readings of both beams and the corrector settings. (2) At each step form the averaged reading changes Δy_b of both beams, adding back the known corrector effect. (3) Estimate the misalignment drift by

$$\Delta\hat{v} = S^{+}\begin{pmatrix}\Delta y_{\mathrm{CW}}\\ \Delta y_{\mathrm{CCW}}\end{pmatrix}, \qquad S = \begin{pmatrix}R_{\mathrm{CW}} - \rho I\\ R_{\mathrm{CCW}} - \rho I\end{pmatrix}, \tag{4}$$

with the discrepancy-principle threshold; ρ describes the BPM mounting (Sec. V A). (4) Apply $\Delta\hat{v}$ with quadrupole-mounted dipole correctors (Sec. V C). (5) Repeat BBA when the accumulated symmetric drift reaches a threshold. Static BPM offsets enter both beams alike, i.e. the sum channel, and are degenerate with the symmetric misalignment (a 100 μm rms offset without reference gives $\sim 200\times$ target); they cancel in the differences of step 2, leaving only the change δ of the BPM center relative to the quadrupole center since the last BBA.

Performance. For $\sigma = 10$ μm, 14 nm noise and a fixed threshold of 10^{-2} (Fig. 4), the observable $C = \frac{1}{2}|f_{\mathrm{CW}} - f_{\mathrm{CCW}}|$ (the EDM reverses with the beam) has a median of 49× target uncorrected, 12 after single-beam and 0.0015 (numerical floor) after two-beam correction. With a lower threshold the single beam also reaches the floor (Table II): at fixed monitors the advantage is mostly conditioning; the decisive advantage appears when monitors move (Sec. V A). The scheme does not require quadrupole polarity reversal, which could save the second storage per measurement.

V. DOMAIN OF VALIDITY

Errors that enter the sum channel compete with the symmetric component. Unless stated otherwise: 5 μm rms symmetric drift after BBA, 14 nm noise, ten patterns, C in units of the target.

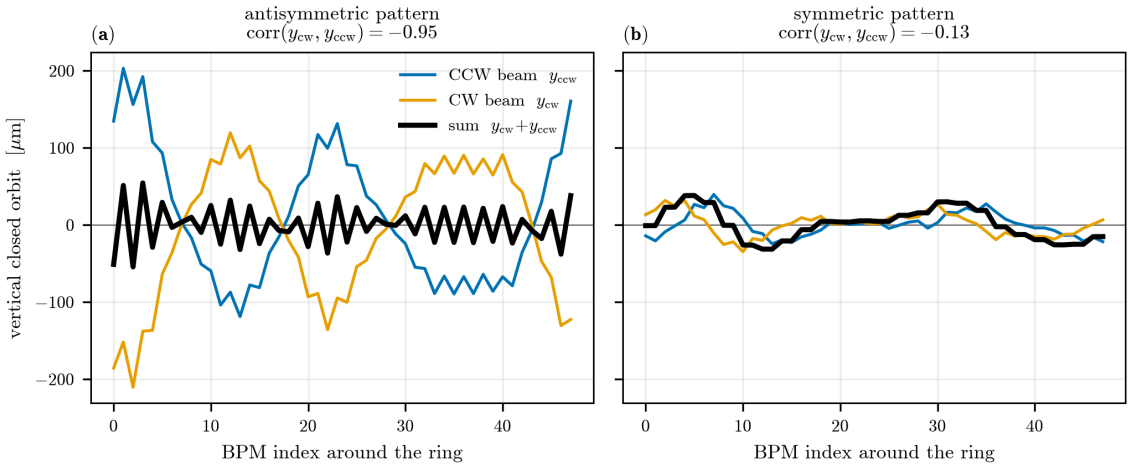


FIG. 3. Vertical closed orbits of the two beams (10 μm rms). (a) Antisymmetric pattern: nearly opposite orbits, small sum (black). (b) Symmetric pattern: nearly equal orbits, the sum survives.

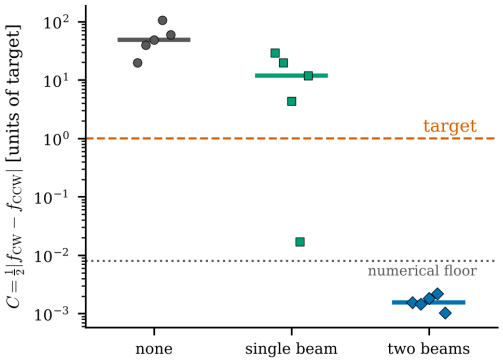


FIG. 4. False EDM at $\sigma = 10$ μm (five patterns; bars: median) without correction, after single-beam and after two-beam correction (fixed monitors, threshold 10^{-2}). Dotted: numerical floor.

A. Monitor model

A BPM measures the beam relative to its electrical center. We write the change of the reading of beam b as $\Delta r_b = R_b \Delta v - u + n_b$ with $u = \rho \Delta v + \delta$: $\rho = 0$ for a monitor fixed in the laboratory, $\rho = 1$ for one rigidly attached to its quadrupole, δ the relative BPM–quadrupole drift. Long-wavelength ground motion moves quadrupoles and nearby monitors together ($\rho \to 1$) [17]. A rigidly attached BPM reads the beam relative to the quadrupole center, the quantity that drives the false EDM. We compare single-beam fits assuming fixed monitors (E1) or using $(R - \rho I)$ (E2) and two-beam fits assuming fixed monitors (E3) or using Eq. (4) (E4) (Table II, Fig. 5).

With fixed monitors both fits reach the floor; the two-beam advantage in residual (threefold) drops to 1.2 with a 0.2% gradient error not included in the model. If monitors follow the quadrupoles but the estimator assumes fixed monitors (E3), most of the symmetric drift survives, yet C stays below target (maximum 0.33): the fit removes 76–78% of the power at $m = 2$–3, where the spin is most sensitive, and leaves harmonics to which it is weakly sensitive. The margin is only three to five. Estimators that know the mounting (E2, E4) reach the floor; E4 tolerates a 10% error in ρ, whereas E2 exceeds the target for one of six patterns at $\hat{\rho} = 0.9$.

TABLE II. Residual symmetric misalignment (200 patterns, median) and false EDM C (10 patterns, median and maximum) after correcting a 5 μm symmetric drift. Uncorrected: $C = 6.3$.

ρ	estimator	residual [μm]	C median	C max
0	E1	0.051	0.008	0.035
0	E3	0.017	0.005	0.012
0.5	E3	3.30	0.013	0.05
1	E1	4.88	3.5	19
1	E2	0.115	0.006	0.017
1	E3	4.39	0.19	0.33
1	E4	0.012	0.003	0.011

With rigidly attached monitors the remaining error is δ [Fig. 5(b)]: the two-beam result stays at the floor up to $\sigma_\delta = 1$ μm, while the single-beam one exceeds the target in 7 of 8 patterns at 1 μm. The two-beam reconstruction error ($1.1\,\sigma_\delta$) lies to 93% in a uniform translation and tilt of the ring ($m = 0, 1$), near-null directions of $(R - I)$ to which the spin is insensitive, since the beam follows a rigid displacement and still passes through the quadrupole centers. BPMs mounted in or on quadrupoles are common in linacs [19, 20]; light sources separate them because of synchrotron-radiation heating, absent here. In this ring a pickup can be placed inside the quadrupole aperture or on the quadrupole yoke; BPM electronics drifts of 0.1 μm/h [21] and ≤ 1 μm per week [22] are spec-

TABLE III. Channel decomposition (units of target). "Common trim": equal roll correction of all deflectors (mean roll removed).

Configuration	even S_+	odd S_-
EDM (10^{-29} e·cm)	0.004	0.986
EDM + random rolls (3×10^{-7})	-1.26×10^{8}	1.02
EDM + same rolls + common trim	0.005	0.992
Uniform radial B (10^{-11} T)	0.017	0.006

ified in modern light sources, consistent with $\sigma_\delta \lesssim 1$ μm between calibrations.

B. Electrostatic deflector errors

A relative field error ε deflects the beam horizontally; a roll ϕ adds a vertical field $E_y \approx E_0\phi$. In a single deflector $\varepsilon = 10^{-4}$ and $\phi = 0.1$ mrad distort the closed orbit by 418 and 525 μm. Since the electric force does not reverse with the beam, these orbits are the same for both beams and fall in the sum channel.

Orbit. A fit with quadrupole columns only mistakes a deflector drift for a symmetric misalignment (≈ 1 μm per 10^{-7} in ε or 0.1 μrad in ϕ) [Fig. 6(a)]. Adding 48 deflector columns per plane keeps the problem full rank but raises the condition number from 73 (118) to 5.6×10^3 (7.9×10^3), because a few deflector and quadrupole patterns produce nearly the same orbit near the tune. The leakage disappears (0.036 μm with an exact model, 0.4 μm at $\varepsilon = 10^{-5}$ with a 0.2% gradient error) at the cost of a doubled noise amplification.

Spin, first order. A roll gives a vertical electric kick $\theta_E = (2\pi/48)\phi$, which the closed orbit balances by quadrupole kicks, $\sum\theta_B = -\sum\theta_E$; the magnetic field rotates the spin $G\gamma$ times more than the momentum, so $f = G\gamma(2\pi/48)\sum_i \phi_i/T_{\rm rev}$ ($1.26\times10^8\times$ target for 3×10^{-7} rms rolls). It depends only on the mean roll: removing the mean of seven random roll patterns brings f to the floor ($|f| \le 0.022$). It has the same sign for both beams, i.e. it lives in the even channel $S_+ = \frac{1}{2}(f_{\rm CW} + f_{\rm CCW})$, while the EDM lives in the odd channel $S_- = \pm C$ (Table III). A uniform radial magnetic field has no first-order effect, since both the field and the restoring quadrupole kicks are magnetic.

In a real ring the two beams are not exactly symmetric—for instance they are frozen at slightly different energies; the roll term depends on energy as $d\ln f/d\delta \approx -2.5$ (CW) and -2.9 (CCW), so the leak into the odd channel is $C \approx |S_+|\cdot\epsilon$ with $\epsilon \approx 2.7\,\Delta\delta$. Uncompensated, this would demand CW/CCW symmetry at the 10^{-8} level. The baseline design already removes S_+: Ref. [2] (Sec. III B and Table IV) applies a common vertical electric trim that keeps the same-sign vertical precession of both beams below 10^{-6} rad/s. Table III confirms that this trim leaves the EDM untouched, under the conditions studied. The leak formula gives the requirement on it: for $\Delta\delta = 10^{-7}$, S_+ must stay below $\sim 10^{-4}$ rad/s for $C \lesssim 0.03$, so the 10^{-6} rad/s of Ref. [2] leaves a leak $C \sim 3\times10^{-4}$ and tolerates $\Delta\delta$ up to $\sim 10^{-5}$.

Spin, second order, and individual trims. Zero-mean rolls still produce a vertical closed orbit, whose product with the horizontal orbit generates a geometric phase in the odd channel: with 0.3 μm quadrupole misalignments, zero-mean rolls raise C from 0.13 to 1.55. The common trim cannot remove this; individual deflector trims, using the deflector part of the fit (field by the deflector voltage, roll by an up–down electrode voltage difference), can. Individual deflector errors are poorly estimated (25–80% remains), but the poorly estimated directions have little orbit effect: the deflector-induced orbit drops by 97–99% and, with field and roll drifts of 3×10^{-7} and a 5 μm symmetric quadrupole drift, $C = 0.010$ (0.016 with a 0.2% gradient error) against 0.75 with quadrupole correction alone. In a matched test (five patterns, 0.3 μm quadrupole misalignment, 3×10^{-7} rms rolls) the common trim alone leaves $C = 1.7$ (median), while common plus individual trims, with the quadrupoles corrected by the same fit, give $C = 0.010$ (maximum 0.019).Figure 6(b) scans the deflector drift: with an exact model the trimmed result stays at the floor up to 10^{-5}; with a 0.2% gradient error it reaches the target at 3–5$\times10^{-6}$. The tolerable drift between trims is set by the accuracy of the response model. Present high-voltage supplies quote 10^{-5} ripple [23] and 10^{-4} long-term stability [24]; common-mode changes are absorbed by the frozen-spin feedback, and only independent slow drifts between trims matter.

C. Corrector type

A dipole winding in a quadrupole adds a uniform field ΔB and shifts the magnetic center by $\Delta B/g$; with $\Delta B = g\Delta\hat{v}$ the quadrupole acts as aligned for both beams and the spin (movers are equivalent). For $g = 0.2$ T/m and drifts of a few microns, $\Delta B \sim 1$ μT, a small trim winding. Correctors at the deflectors fail: magnetic ones act only on the difference channel, whereas a symmetric drift needs a same-sign displacement of both beams; electrostatic ones can steer both beams through the quadrupole centers, but the needed kicks (5–8$\times10^{-6}$ in ε, ϕ) generate the deflector effects above ($C = 1.3$–7.6).

D. Correlated misalignment

Survey errors and ground motion correlate misalignments along the ring. In a periodic random walk, $\langle(v_i - v_j)^2\rangle \propto |s_i - s_j|$—the spatial structure of the ATL law [18], one correlation scenario among others—the symmetric component carries 84% of the variance (symmetric/antisymmetric rms 2.3), against about half for uncorrelated errors. The two-beam correction is linear and its residual does not depend on the pattern statistics. If the ground also moves the monitors, the case is $\rho \approx 1$

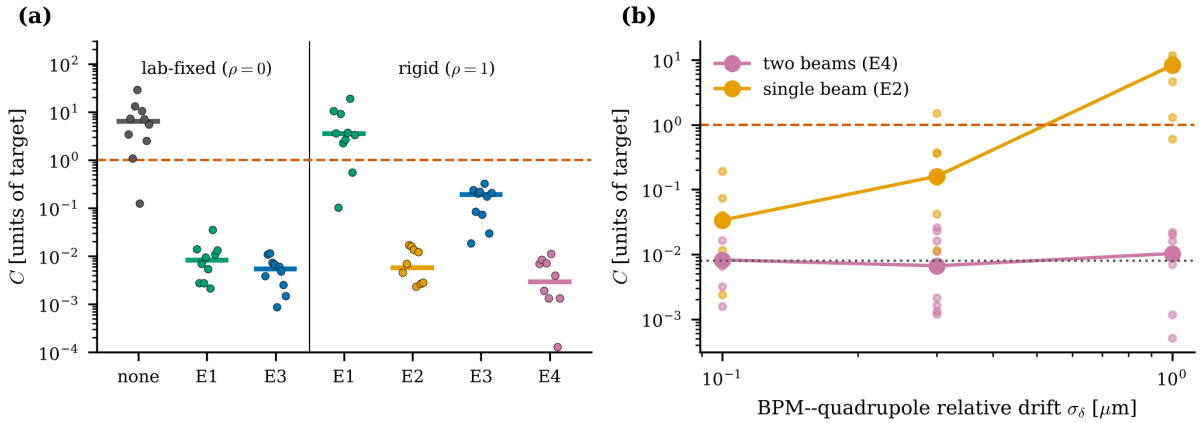


FIG. 5. (a) False EDM after correcting a 5 μm symmetric drift, for fixed ($\rho = 0$) and rigidly attached ($\rho = 1$) monitors and the estimators of Table II. (b) Rigidly attached monitors with relative BPM–quadrupole drift σ_δ. Dashed: target; dotted: floor.

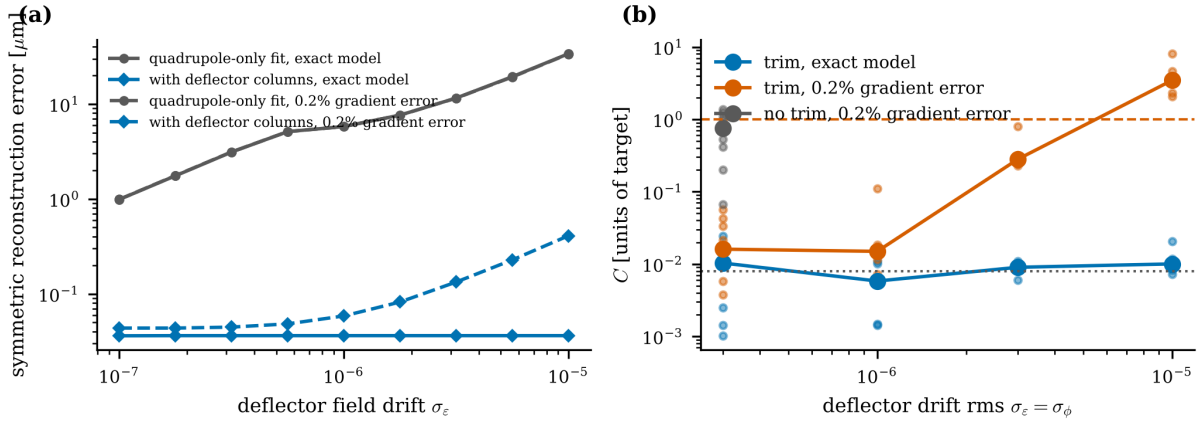


FIG. 6. (a) Error of the reconstructed symmetric quadrupole drift versus deflector field drift, with quadrupole-only and with deflector columns. (b) False EDM after quadrupole correction and deflector trim versus the deflector drift amplitude.

of Sec. V A: a fixed-monitor estimator recovers about a third of the symmetric drift, an estimator with $\rho = 1$ recovers it to 0.01 μm.

VI. OPERATING SCENARIO

We followed the machine for one week in 1 h steps: ATL ground motion with $A = 10^{-5}$ $\mu\text{m}^2/(\text{s m})$, close to the value measured at LEP [18]; 0.3 μm rms residual after BBA; BPM–quadrupole drift 0.2 $\mu\text{m}/\sqrt{\text{day}}$, with BBA repeated when its expected value reaches 0.3 μm ($\approx$ 54 h); BPM readings with 10 μm turn-by-turn resolution, 60 s averaging and a 10 nm floor. Two-beam monitoring with rigid monitors keeps the false EDM at the level set by the BBA residual (median $C = 0.05$–0.09, maximum 0.18 over 15 values) (Fig. 7). Single-beam monitoring and the fixed-monitor model hover around the target (median up to 1.2 and 1.3); periodic BBA alone rises to $C \approx 45$ between calibrations. The monitor detects a symmetric drift of $3k\sigma_n \approx 0.02$ μm ($k = 0.75$ the noise gain, σ_n the BPM noise per reading). Deflector drifts are not included in this scenario, and the spin results rest on five patterns.

VII. CONCLUSIONS

The symmetric component of quadrupole misalignment is four to seven times less visible to the closed orbit at low harmonics, because of the β modulation between QF and QD, while its weight in the false EDM is reduced less; single-beam correction leaves it largely in place. The two counter-rotating beams separate the components, the symmetric one appearing in the sum of the orbits. Because static offsets are degenerate with it, the scheme is a drift monitor between BBA calibrations and needs no quadrupole polarity reversal. Under the conditions studied it requires: a monitor model that matches the mounting (rigidly attached monitors with a model that

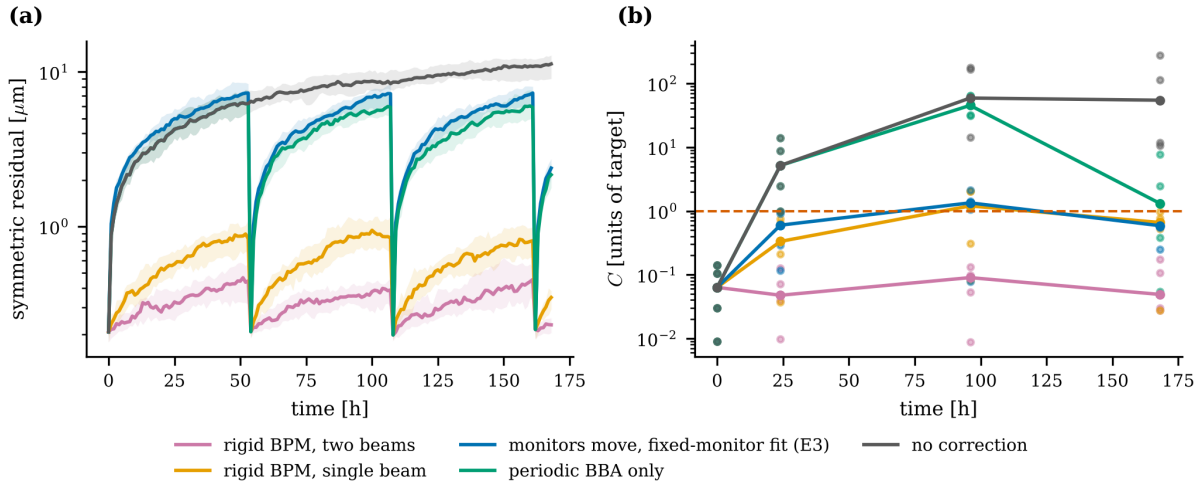


FIG. 7. One-week scenario. (a) Residual symmetric misalignment (20 patterns, median and 16–84% band). (b) False EDM from tracking (5 patterns; lines: median). BBA at 54, 108 and 162 h.

knows it tolerate BPM–quadrupole drifts up to at least 1 μm); deflector columns in the fit and individual deflector trims (tolerable drift between trims $\sim 3 \times 10^{-6}$ with a 0.2% gradient error); the common vertical trim of Ref. [2], which we find removes the first-order roll effect without touching the EDM channel; and quadrupole-mounted correctors. Under these conditions two-beam monitoring keeps the false EDM well below target between calibrations over a simulated week. Finally, the false EDM of a misaligned lattice must be evaluated at the frozen-spin energy; a fixed-momentum evaluation depends on the launch point.

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Appendix A: Launch-point dependence

The difference between the fixed-momentum and frozen-spin evaluations grows with the closed-orbit amplitude at launch. For the patterns of Fig. 4 it is $\leq 0.02\times$ target after two-beam correction (rms orbit product $\approx 0.01\ \mu\mathrm{m}^2$), ≤ 0.2 for the corrected beam after single-beam correction ($\approx 1\ \mu\mathrm{m}^2$) and 1–3000 for uncorrected orbits (10^2–$10^4\ \mu\mathrm{m}^2$); it differs between the beam directions and does not cancel in C. The four frozen-spin solutions of Sec. II are the same orbit: their cell-to-cell momentum differences agree to four digits with those measured along one closed orbit. The frozen-spin energy is found independently of the starting rule of the interpolation; if the second point is far from $\omega_h = 0$ ($|\omega_1| > 10^{-3}$ rad/s), a third measurement at the interpolated energy is taken.

Appendix B: Gradient modulation: breathing and conditioning

Changing one quadrupole gradient changes the beta functions of the whole ring and every response-matrix element; with all quadrupoles modulated these changes dominate the reconstruction [Fig. 8(a)]. The orbit difference between two gradient settings cancels static offsets, but $\Delta R \approx \epsilon_g \partial R/\partial g$, so $\|\Delta R^{-1}\| \approx \|R^{-1}\|/\epsilon_g$ (condition number 3.7×10^4); the inversion fails selectively for the symmetric directions [Fig. 8(b)].

Appendix C: Two-dimensional illustration of the sum

Let $v = (a, s)^\top$, $R_{\mathrm{CW}} = \mathrm{diag}(G, g)$ with $G \gg g$, $R_{\mathrm{CCW}} = \mathrm{diag}(-G, g)$, and a static offset $b = (b_a, b_s)^\top$ common to both beams. Then $y_{\mathrm{CW}} - y_{\mathrm{CCW}} = (2Ga, 0)^\top$ and $y_{\mathrm{CW}} + y_{\mathrm{CCW}} = (2b_a, 2gs + 2b_s)^\top$: the difference gives a, free of offsets; the sum gives s only together with b_s, through the small gain g. The offset cancels in the change of the sum, $\Delta s = \Delta(y_{\mathrm{CW}} + y_{\mathrm{CCW}})/2g$, which makes the scheme a drift monitor.

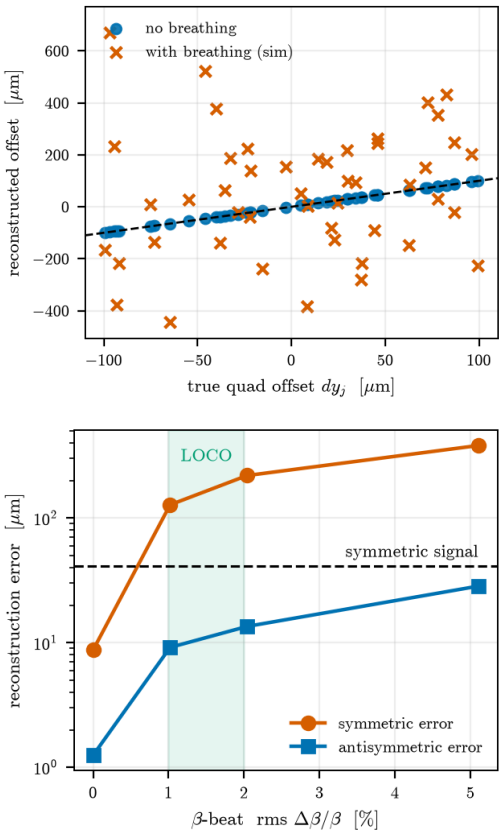


FIG. 8. (a) Reconstruction with all quadrupoles modulated, with and without breathing. (b) Reconstruction error of the orbit-difference approach versus β-beat.

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